TECHNICAL PRINCIPLES

5.2 What materials should be selected when designing and manufacturing products and prototypes in product design?

a. Understand that most products consist of multiple materials and that product designers are required to discriminate between them appropriately for their use, including:

- i. hardwoods and softwoods, such as:
  - oak, teak and beech; pine, spruce and fir
- ii. manufactured boards, such as:
  - plywood, MDF and block board
- iii. ferrous and non-ferrous metals, such as:
  - cast iron, mild steel and stainless steel; aluminum and copper
- iv. metal alloys, such as:
  - brass, bronze and tungsten
- v. thermopolymers and thermosetting polymers, such as:
  - PET, HDPE, PVC, LDPE, polypropylene, polystyrene and ABS; urea formaldehyde, epoxy resin and polyester resin.
- vi. natural and synthetic fibres, such as:
  - cotton, wool and silk; polyester and nylon
- vii. textile fabrics, such as:
  - woven, non-woven, knitted and blended textiles
- viii. composite materials, such as:
  - fibre-reinforced plastics, glass-reinforced plastics (GRP) and carbon fibre (CFRP)
- ix. modern materials, such as:
  - e-textiles, super-alloys, graphene, bioplastics and nanomaterials
- x. smart materials, such as:
  - thermochromic, photochromic and electrochromic materials; shape memory alloy and shape memory polymers; conductive paints and e-textiles.